

Question 1 (2 points)

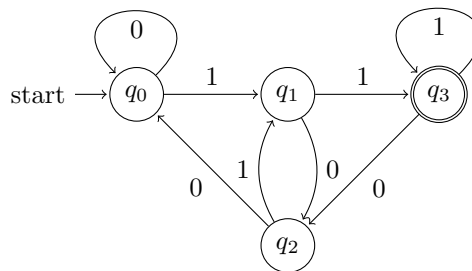
Let L be a regular language. Then according to the pumping lemma for regular languages, there exists an integer $p \geq 1$ such that every string $w \in L$ can be written as $w = xyz$ satisfying the following conditions:

- $|y| \geq 1$
- $|xy| \leq p$ and
 - A. $\forall n \geq 0, xy^n z \in L$
 - B. $\exists n \geq 0, xy^n z \in L$
 - C. $\forall n \geq 0, xyz^n \in L$
 - D. $\exists n \geq 0, x^n yz \in L$

Option A is correct. Refer to the Pumping Lemma (Section 6.1 in Week 3 Lecture Notes)

Question 2 (2 points)

Which states are equivalent in the following DFA?



- A. q_0 and q_1 only
- B. q_1 and q_2 only
- C. q_0 and q_2 only**
- D. q_0, q_1 and q_2

Option C is correct. Minimize the given DFA according to the process given in Section 7.1 in Week 3 Lecture Notes. You will find that the states q_0 and q_2 belong to the same equivalence class at the end of the process, but q_1 does not. That is, $q_0 \approx q_2$.

Question 3 (2 points)

Which of the following is a regular language?

- A. $\{0^m 1^m \mid m > 0 \text{ is a natural number}\}$
- B. $\{0^m 1^n \mid m, n > 0 \text{ are natural numbers}\}$**
- C. $\{0^m 1^m \mid m \geq 0 \text{ is a natural number}\}$
- D. $\{0^m 1^n \mid m, n \geq 0 \text{ are natural numbers such that } m \geq n\}$

Option B is correct. The regular expression corresponding to this language is 00^*11^* . It can be shown using the pumping lemma that the other languages are not regular.

Question 4 (2 points)

Which of the following is not a regular language?

- A. $\{0^m \mid m \text{ is a natural number}\}$
- B. $\{0^{2m} \mid m \text{ is a natural number}\}$
- C. $\{0^{3m+2} \mid m \text{ is a natural number } m \geq n\}$

D. $\{0^{m^2} \mid m \text{ is a natural number}\}$

Option D is correct. We can use pumping lemma to show that this language is not regular.

We will prove this via contradiction. Let's say this language is regular. Then it must satisfy the pumping lemma. Therefore there exists an integer $p \geq 1$ such that for some $k \leq p$, 0^{m^2+k} also belongs to the language. (The substring y in the pumping lemma has to be some string of 0's of length $k \leq p$ here and when we pump that substring y once into the string 0^{m^2} , we get 0^{m^2+k}). But for $m = p$, $m^2 + k \leq p^2 + p \leq (p+1)^2$. Thus, $m^2 + k$ is not the square of a natural number as it lies between the squares of two consecutive natural numbers p and $p+1$. This is a contradiction.

Question 5 (2 points)

Which of the following statements is not true?

- A. Any regular language satisfies the pumping lemma
- B. Any language that does not satisfy the pumping lemma cannot be regular
- C. Any language that satisfies the pumping lemma must be regular.**
- D. A language that is not regular may also satisfy the pumping lemma.

Option C is correct. Refer to the Pumping Lemma (Section 6.1 in Week 3 Lecture Notes)

Question 6 (2 points)

Which of the following languages is not regular?

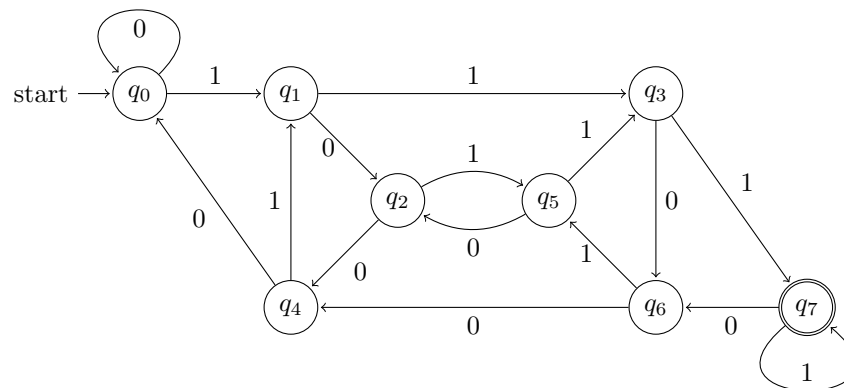
- A. $\{0^n \mid n \text{ is prime}\}$**
- B. $\{0^i 1^j \mid i, j \geq 0\}$
- C. Set of binary strings where every occurrence of 1 is followed by 0
- D. Set of binary strings such that the numbers represented by them in binary representation is divisible by 3

Option A is correct. We can use pumping lemma to show that this language is not regular.

We will prove this via contradiction. Let's say this language is regular. Then it must satisfy the pumping lemma. Therefore there exists an integer $p \geq 1$ such that for some $k \leq p$, 0^{n+kn} also belongs to the language. (The substring y in the pumping lemma has to be some string of 0's of length $k \leq p$ here and when we pump that substring y n times into the string 0^n , we get 0^{n+kn}). $n + kn = n(k+1)$ is not a prime number. This is a contradiction.

Question 7 (2 points)

How many states are there in the minimal DFA corresponding to the following DFA?



- A. 3

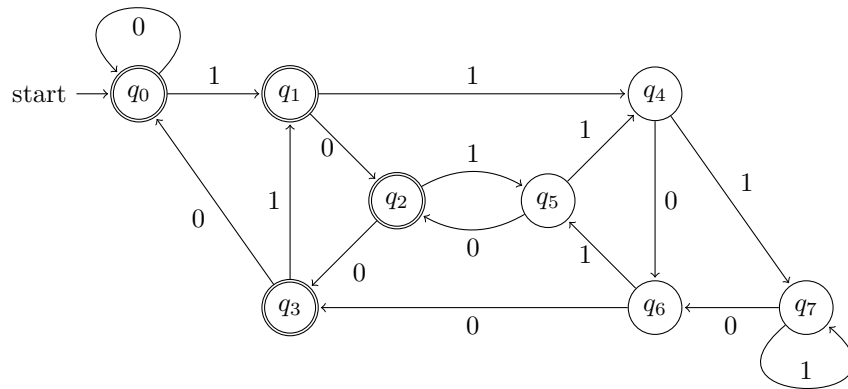
-
- B. 4
C. 6
D. 8

Option B is correct.

Minimize the given DFA according to the process given in Section 7.1 in Week 3 Lecture Notes. You will find that there are four equivalence classes of states for this DFA $\{q_0, q_2, q_4, q_6\}$, $\{q_1, q_5\}$, $\{q_3\}$ and $\{q_7\}$. Therefore, we can construct a minimal DFA with 4 states corresponding to these 4 equivalence classes.

Question 8 (2 points)

How many final/accepting states are there in the minimal DFA corresponding to the following DFA?



- A. 1
B. 2
C. 3
D. 4

Option C is correct.

Minimize the given DFA according to the process given in Section 7.1 in Week 3 Lecture Notes. You will find that there are six equivalence classes of states for this DFA $\{q_0, q_3\}$, $\{q_1\}$, $\{q_2\}$, $\{q_4, q_7\}$, $\{q_5\}$ and $\{q_6\}$. Out of these, three correspond to accept states- $\{q_0, q_3\}$, $\{q_1\}$ and $\{q_2\}$. Therefore, in the minimal DFA, there will be three accept states corresponding to these three equivalence classes.